

An exact exponent obstruction for lower-end projection-cone simplex stability

Research supplement to entry005; AI-assisted research note

7 October 2026

Outcome. In every fixed dimension $d \geq 3$, any power estimate for the lower-end defect of entry005, whether in Banach–Mazur distance to any simplex or in centroid containment by every maximum-volume inscribed simplex, must have exponent at most $1/(d-1)$. The example also gives an exact lower bound on the constant in the latter estimate at this exponent. This is a necessary limit, not a claim that the endpoint exponent has been proved for arbitrary bodies. The existing explicit power theorem in the quantitative supplement is not reproved here.

1 Conventions and public-source scope

For a full-dimensional convex body in $\mathbb{R}^d$, use exactly the public entry005 convention

$$R(K) = \frac{|\Pi K|}{|K|^{d-1}}, \quad a(K) = \left(\frac{d}{d+1} \right)^d \frac{R(\mathcal{P}K)}{R(K)} - 1, \quad \mathfrak{e}(K) = a(K) - \frac{1}{d+1}.$$

Here $\mathcal{P}K$ is a pyramid over K and ΠK is its projection body. The definition and the determinant formula below were read in public v3; v4 preserves the definition. The latest public v5 at commit 31e3d8a37e4a3051a7f5a2535be1c75642e15bcc concerns balanced product/join recursions and does not replace the lower-end definition or add a quantitative lower-end theorem within that versioned package.

The additive quantitative supplement is available publicly at commit 6785c1c830f8e19e2eb07b0bb89f4d475a8b154a. Its source defines a and E exactly as above and proves the every-maximum-simplex bound with exponent $1/[d(d^2 + 2d + 2)]$. Its stated inputs are the v2–v3 invariant identities and classical Cauchy and mixed-volume formulas; its manifest pins the audited dependency commit dd969ea280508a15995fbe83a80cac0f6d227f73. This note does not repeat that quantitative proof.

For a simplex $S \subset K$ with centroid g_S , define the *excess* centroid dilation by

$$E(K, S) = \inf\{\epsilon \geq 0 : K \subset g_S + (1 + \epsilon)(S - g_S)\}.$$

Thus the dilation factor itself is $1 + E$. Write $M(K)$ for the maximum volume of an inscribed d -simplex. For any fixed reference simplex Δ_d , the usual affine Banach–Mazur distance to its affine class can equivalently be written

$$d_{\text{BM}}(K, \Delta_d) = \inf\{\lambda \geq 1 : S \subset K \subset v + \lambda S, \ S \text{ a full-dimensional simplex, } v \in \mathbb{R}^d\}. \quad (1)$$

The free translation in (1) and optimization over all S are important. They are both absent from the prescribed-centroid, prescribed-maximum-simplex quantity E .

For a polytope with facet area times outward unit normal u_i and facet area times support number b_i , public v3 gives

$$a(K) = \frac{\sum_{|I|=d+1} |\det((u_i, b_i) : i \in I)|}{d|K| \sum_{|J|=d} |\det(u_j : j \in J)|}. \quad (2)$$

It is legitimate to use (2) before translating an interior point to zero: a translation by z replaces b_i by $b_i - u_i \cdot z$, an elementary row operation in every lifted determinant. No nonpositive b_i is used as a probability weight in this note.

2 The obstruction family and its complete calculation

Let $e_1, \dots, e_d$ be the coordinate vectors and, for $0 < t < 1$, set

$$K_t = \{x \in \mathbb{R}^d : x_i \geq 0, t \leq x_1 + \dots + x_d \leq 1\} = \text{conv}\{e_1, \dots, e_d, te_1, \dots, te_d\}. \quad (3)$$

This is a standard simplex with a homothetic cap cut from one vertex.

Theorem 1. *For every integer $d \geq 2$ and $0 < t < 1$, the following assertions hold.*

(i) *With $q = t^{d-1}$, the invariant is exactly*

$$a(K_t) = \frac{1 + (d-1)q - (d-1)t^d - qt^d}{(1-t^d)(d+1+(d-1)q)}, \quad (4)$$

and consequently

$$\mathfrak{e}(K_t) = \frac{t^{d-1}[d(d-1) - (d+1)(d-2)t - 2t^d]}{(d+1)(1-t^d)(d+1+(d-1)t^{d-1})}. \quad (5)$$

In particular,

$$\lim_{t \downarrow 0} \frac{\mathfrak{e}(K_t)}{t^{d-1}} = c_d := \frac{d(d-1)}{(d+1)^2} > 0. \quad (6)$$

(ii) *The maximum simplex volume is*

$$M(K_t) = \frac{1-t}{d!}. \quad (7)$$

Every maximum-volume inscribed simplex, including those with vertices in the relative interiors of faces, has the form

$$S_p = \text{conv}\{e_1, \dots, e_d, tp\}, \quad p_i \geq 0, \quad \sum_{i=1}^d p_i = 1. \quad (8)$$

For each such simplex,

$$E(K_t, S_p) = (d+1)t \max_i p_i. \quad (9)$$

Thus

$$\inf_{S \text{ maximum}} E(K_t, S) = \frac{d+1}{d}t, \quad \sup_{S \text{ maximum}} E(K_t, S) = (d+1)t. \quad (10)$$

(iii) *The distance to any simplex satisfies*

$$\left(1 + t + \dots + t^{d-1}\right)^{1/d} \leq d_{\text{BM}}(K_t, \Delta_d) \leq 1 + t. \quad (11)$$

In fact, for every fixed maximum simplex S_p , its smallest enclosing translated homothet has factor exactly $1 + t$. In particular, for $0 < t < 1$,

$$\frac{t}{2d} \leq d_{\text{BM}}(K_t, \Delta_d) - 1 \leq t, \quad \liminf_{t \downarrow 0} \frac{d_{\text{BM}}(K_t, \Delta_d) - 1}{t} \geq \frac{1}{d}. \quad (12)$$

(iv) *The Minkowski asymmetry is exactly*

$$s(K_t) := \inf\{r \geq 1 : \exists z, -(K_t - z) \subset r(K_t - z)\} = d - t. \quad (13)$$

Proof. We prove each assertion directly.

Facet determinants. Put $c = 1/(d-1)!$ and $r = 1 - q$. The d coordinate facets have area $c(1 - t^{d-1})$. The top and bottom facets have areas $c\sqrt{d}$ and $c\sqrt{d}t^{d-1}$, respectively. The facet data in the coordinates (3) are therefore

$$(u_i, b_i) = (-cre_i, 0) \quad (1 \leq i \leq d), \quad (u_+, b_+) = (c\mathbf{1}, c), \quad (u_-, b_-) = (-cq\mathbf{1}, -ct^d). \quad (14)$$

The volume is $|K_t| = (1 - t^d)/d!$, so $d|K_t| = c(1 - t^d)$.

For horizontal d -minors, the all-coordinate subset contributes $c^d r^d$; the d subsets with the top facet contribute $c^d r^{d-1}$ each; and the d subsets with the bottom facet contribute $c^d q r^{d-1}$ each. Subsets with both top and bottom have determinant zero because their normals are parallel. Hence the horizontal sum is

$$H = c^d r^{d-1} [d + 1 + (d-1)q]. \quad (15)$$

For the lifted $(d+1)$ -minors, the subset containing all coordinate facets and the top facet contributes $c^{d+1} r^d$. Replacing the top facet by the bottom facet gives $c^{d+1} r^d t^d$. Each of the d subsets with top, bottom and $d-1$ coordinate facets contributes $c^{d+1} r^{d-1} (q - t^d)$; here $q - t^d = q(1 - t) > 0$. Thus the lifted sum is

$$L = c^{d+1} r^{d-1} [1 + (d-1)q - (d-1)t^d - qt^d]. \quad (16)$$

Dividing L by $d|K_t|H$ gives (4); subtracting $1/(d+1)$ gives (5). The bracket in (5) decreases strictly to zero as $t \uparrow 1$, so the defect is positive. Dividing (5) by t^{d-1} and passing to $t = 0$ proves the exact limit (6), without any numerical fitting.

Maximum simplices, including nonvertex ones. First consider a simplex selected from the $2d$ vertices in (3). If a ray $\mathbb{R}_+ e_i$ is unrepresented, the simplex lies in the coordinate hyperplane $x_i = 0$ and has zero volume. Otherwise $d+1$ distinct vertices on d rays give exactly one doubled ray, with vertices e_j, te_j , and one vertex on every other ray. Subtract the two doubled columns in the lifted vertex determinant. If k of the other $d-1$ selected vertices are bottom vertices, its absolute determinant is

$$(1-t)t^k. \quad (17)$$

The largest value is $1-t$, attained precisely by all d top vertices and one bottom vertex. Repeated vertices give determinant zero.

Now let $z_0, \dots, z_d$ be arbitrary vertices of an inscribed simplex. Represent each z_j as a convex combination of the $2d$ vertices, and independently sample a vertex Y_j from that combination. Multiaffinity of the lifted determinant D gives

$$|D(z_0, \dots, z_d)| = |\mathbb{E}D(Y_0, \dots, Y_d)| \leq \mathbb{E}|D(Y_0, \dots, Y_d)| \leq 1-t. \quad (18)$$

This proves the upper bound (7), already attained by the preceding examples. If the simplex is maximum, equality in (18) implies that $|D(Y_0, \dots, Y_d)| = 1-t$ almost surely. By (17), exactly one of the Y_j is a bottom vertex and the other d are the d distinct top vertices, almost surely.

Let I_j be the indicator that Y_j is a bottom vertex. Independence and $\sum_j I_j = 1$ almost surely imply

$$0 = \text{Var}\left(\sum_j I_j\right) = \sum_j \mathbb{P}(I_j = 1)[1 - \mathbb{P}(I_j = 1)].$$

Thus every slot has deterministic top/bottom type, with precisely one bottom slot. The supports of the d top slots must be pairwise disjoint; otherwise independent sampling gives a repeated top vertex with positive probability. There are d nonempty disjoint supports in a set of d top vertices, so each is a singleton. The bottom slot is any convex combination tp . This proves (8). Conversely, the height above the hyperplane $\sum x_i = 1$ of any point tp is fixed, so every simplex (8) has volume $(1 - t)/d!$.

Centroid containment. Write $s_x = \sum_i x_i$. Barycentric coordinates for S_p , with index 0 for tp , are

$$\beta_0(x) = \frac{1 - s_x}{1 - t}, \quad \beta_i(x) = x_i - tp_i \beta_0(x) \quad (1 \leq i \leq d). \quad (19)$$

Evaluation at the $2d$ vertices of K_t shows

$$\min_{K_t} \beta_0 = 0, \quad \min_{K_t} \beta_i = -tp_i. \quad (20)$$

For the latter minimum, use te_j with $j \neq i$, available since $d \geq 2$. Affine functions attain their extrema on these vertices. The centroid-dilated simplex $gs_p + (1 + \epsilon)(S_p - gs_p)$ is exactly the region $\beta_i \geq -\epsilon/(d + 1)$ for every $0 \leq i \leq d$. Equations (20) give (9), and optimizing $\max_i p_i$ on the probability simplex gives (10).

Translated containment and distance to any simplex. For barycentric coordinates of any simplex S , a translated homothet $v + \lambda S$ has inequalities $\beta_i \geq a_i$ with $\sum_i a_i = 1 - \lambda$. Conversely any such thresholds describe a translated homothet. To contain K , one needs $a_i \leq \min_K \beta_i$ for each i , so

$$\lambda \geq 1 - \sum_i \min_K \beta_i. \quad (21)$$

Choosing all thresholds equal to their minima realizes equality whenever $S \subset K$. For (20), the sum is $-t$, giving factor exactly $1 + t$. Concretely,

$$K_t \subset -tp + (1 + t)S_p. \quad (22)$$

This proves the upper bound in (11) under the full freedom in (1).

For the lower bound, any simplex $S \subset K_t$ has $|S| \leq M(K_t)$. If $K_t \subset v + \lambda S$, volume monotonicity and (7) imply

$$\lambda^d \geq \frac{|K_t|}{|S|} \geq \frac{|K_t|}{M(K_t)} = \frac{1 - t^d}{1 - t} = 1 + t + \dots + t^{d-1}. \quad (23)$$

Taking the infimum proves the lower bound in (11). Furthermore

$$(1 + t)^{1/d} - 1 = \frac{1}{d} \int_0^t (1 + u)^{-(d-1)/d} du \geq \frac{t}{d 2^{(d-1)/d}} \geq \frac{t}{2d}.$$

Finally the derivative at zero of $(1 + t + \dots + t^{d-1})^{1/d}$ is $1/d$, proving (12). This lower bound concerns *all* simplices, not only maximum simplices. We do not assert an exact formula for the unrestricted distance in (11).

Minkowski asymmetry. The containment in (13) is equivalent to $-K_t + (r+1)z \subset rK_t$. Writing $Z = \sum_i z_i$, its necessary and sufficient facet inequalities are

$$(r+1)z_i \geq 1 \quad (1 \leq i \leq d), \quad 1 + rt \leq (r+1)Z \leq r+t. \quad (24)$$

Summing the coordinate inequalities and using the last upper bound gives $r \geq d-t$. This value is attainable: take $z_i = 1/(d+1-t)$. The bottom inequality follows from

$$d-1 - (d-t)t = (1-t)(d-1-t) \geq 0.$$

All other inequalities in (24) are equalities. This proves (13). $\square$

3 Universal implications and necessary constants

Corollary 2. *Fix $d \geq 3$. A power estimate valid for every full-dimensional body of dimension d , even restricted to a sufficiently small lower-end defect, can have exponent α only if $\alpha \leq 1/(d-1)$, for each of:*

$$d_{\text{BM}}(K, \Delta_d) - 1 \leq C_d \mathfrak{e}(K)^\alpha; \quad (25)$$

$$\inf_{S \text{ maximum}} E(K, S) \leq C_d \mathfrak{e}(K)^\alpha; \quad (26)$$

$$E(K, S) \leq G_d \mathfrak{e}(K)^\alpha \quad \text{for every maximum-volume inscribed simplex } S. \quad (27)$$

At the critical exponent $1/(d-1)$, any constant in (27) must satisfy

$$G_d \geq (d+1) \left(\frac{(d+1)^2}{d(d-1)} \right)^{1/(d-1)} > d+1. \quad (28)$$

Any constant in (26) must be at least $1/d$ times the right-hand side of (28), and any constant in (25) must be at least

$$\frac{1}{d} \left(\frac{(d+1)^2}{d(d-1)} \right)^{1/(d-1)}. \quad (29)$$

These are necessary constants, not claimed optimal universal constants. In particular a dimension-independent constant is impossible in (27) at the critical exponent.

Proof. Equations (6), (10), and (12) give a positive constant times t on the left of each estimate and a constant times $t^{(d-1)\alpha}$ on its right. If $(d-1)\alpha > 1$, division by t and passage to $t \downarrow 0$ is a contradiction. At equality of exponents, take $p = e_1$ in (9) for (28), $p_i = 1/d$ for (26), and the lower-limit bound in (12) for (29). $\square$

Combining the released upper estimate with Corollary 2 places the supremum of admissible powers, for either the every-maximum-simplex quantity or the unrestricted simplex Banach–Mazur distance, in the interval

$$\frac{1}{d(d^2 + 2d + 2)} \leq \alpha_d^* \leq \frac{1}{d-1}. \quad (29a)$$

For $d = 3$ this is $1/51 \leq \alpha_3^* \leq 1/2$. The gap between these two endpoints remains unresolved here.

The tetrahedral case. For $d = 3$, cancellation of $1-t$ in (5) gives the especially simple formula

$$\boxed{\mathfrak{e}(K_t) = \frac{t^2(t^2 + t + 3)}{4(1+t+t^2)(2+t^2)}, \quad \lim_{t \downarrow 0} \frac{\mathfrak{e}(K_t)}{t^2} = \frac{3}{8}.} \quad (30)$$

Every maximum tetrahedron is $S_p = \text{conv}\{e_1, e_2, e_3, tp\}$ and has $E = 4t \max p_i$. Thus no exponent greater than $1/2$ is possible; at exponent $1/2$ the every-maximum-tetrahedron constant is at least $4\sqrt{8/3}$, and the best-maximum-tetrahedron constant is at least $(4/3)\sqrt{8/3}$.

4 A precise obstruction to a direct Rogers–Shephard bridge

Proposition 3. *For the same family in dimension three, put*

$$\delta_{\text{RS}}(K) = 1 - \frac{|K - K|}{20|K|}.$$

Then

$$|K_t - K_t| = \frac{10 - 3t - 6t^2 - t^3}{3}, \quad \delta_{\text{RS}}(K_t) = \frac{3t(1 + 3t)}{10(1 + t + t^2)}. \quad (31)$$

In particular,

$$\lim_{t \downarrow 0} \frac{\delta_{\text{RS}}(K_t)}{t} = \frac{3}{10}, \quad 3 - s(K_t) = t. \quad (32)$$

Neither $\delta_{\text{RS}}(K) \leq C\mathfrak{e}(K)$ nor $3 - s(K) \leq C\mathfrak{e}(K)$ can hold uniformly near the lower endpoint. More generally either of these two bridges with $\mathfrak{e}^\beta$ requires $\beta \leq 1/2$.

Proof. For $x \in \mathbb{R}^3$, let $P = \sum_i (x_i)_+$ and $N = \sum_i (-x_i)_+$. Every representation $x = a - b$ with $a, b \geq 0$ has $a = x_+ + w$, $b = x_- + w$ for some $w \geq 0$. Writing $W = \sum_i w_i$, membership $a, b \in K_t$ is equivalent to

$$\max(0, t - \min(P, N)) \leq W \leq 1 - \max(P, N).$$

Hence

$$K_t - K_t = \{P \leq 1, N \leq 1, |P - N| \leq 1 - t\}. \quad (33)$$

The difference body of the uncut simplex is $\{P \leq 1, N \leq 1\}$. Its two pure-sign orthants each have volume $1/6$, and each of the six mixed orthants has volume $1/2$. Its total volume is therefore $10/3$. The two caps $P - N > 1 - t$ and $N - P > 1 - t$ are disjoint and have equal volumes.

For the first cap, the all-positive orthant contributes $[1 - (1 - t)^3]/6$. Each of the three orthants with two positive coordinates and one negative coordinate contributes

$$\int_{1-t}^1 P(P - 1 + t) dP = \frac{(1 - t)t^2}{2} + \frac{t^3}{3}.$$

Each of the three orthants with one positive coordinate and two negative coordinates contributes

$$\frac{1}{2} \int_{1-t}^1 (P - 1 + t)^2 dP = \frac{t^3}{6}.$$

These integrals follow by summing the nonnegative coordinates: two such coordinates have slice density equal to their sum. The cap volume is consequently $t/2 + t^2 + t^3/6$. Subtracting twice this from $10/3$ gives the first formula in (31). Substitute $|K_t| = (1 - t^3)/6$ and cancel $1 - t$ to obtain the second. Equations (30), (32), and (13) prove the bridge obstructions by taking $t \downarrow 0$. $\square$

5 Literature check, limitations, and verification

Rogers–Shephard and Zhang. Böröczky’s 2005 paper proves linear Banach–Mazur stability for the normalized Rogers–Shephard deficit, and a power $1/d$ estimate for the polar projection (Zhang) deficit. Its Example 19 already uses truncation at all simplex vertices and gives an obstruction beyond $1/(d - 1)$ for that different invariant. Truncation as a sharpness method is therefore prior work. Equations (4)–(10) calculate the entry005 invariant and all its maximum-simplex centroid

distances specifically. Proposition 3 shows why the linear Rogers–Shephard result alone cannot give a linear bound in ϵ in dimension three.

Random simplices. Ambrus–Böröczky’s random-simplex stability paper studies uniform interior sampling; its stability theorem for the simplex as a maximizer is planar. Entry005 instead uses determinants for the centered cone-volume law on the polar boundary. No identity transporting the relevant defects between these sampling models has been established here. The cited theorem is therefore not a quantitative solution for this invariant in $d \geq 3$.

Inverse Minkowski. Böröczky–Machado–Ramos, arXiv:2603.17726v3, prove the sharp Hausdorff exponent $1/(d-1)$ for inverse Minkowski stability in the dual-convex distance of suitably dispersed normalized surface-area measures. This is a plausible conversion step once closeness to a simplex measure has been proved. It does not itself supply a bound on that measure distance from the scalar gap ϵ . This missing implication remains a separate obligation, and no unconditional improved exponent is asserted from it.

Minkowski asymmetry. Brandenberg, González Merino, and Grundbacher (arXiv:2607.27041v2) prove dimension-independent linear-order simplex approximation when $d - s(K)$ is small. Equation (13) identifies that parameter exactly in our family; (6) shows that a linear control of it by ϵ is impossible in $d \geq 3$. Thus this useful geometric theorem also requires a root-type bridge in the present invariant.

What is and is not proved. The exponent ceiling, exact invariant, maximum-simplex classification, centroid excesses, translated excesses for those simplices, asymmetry, and (31) are proved above. The general upper bound with exponent $1/(d-1)$, the exact unrestricted distance in (11), and optimal universal constants are not proved. No first-discovery or best-known claim is made. The exact obstruction is compatible with the quantitative supplement’s much smaller positive exponent.

Exact supporting computation. The accompanying standard-library script `check_exact.py` uses `Fractions.Fraction` throughout. It checks facet determinants, all vertex-subset simplex volumes, barycentric containment for several different maximum simplices, and the dimension-three difference-body volume by independently enumerating supporting facets and triangulating them with rational arithmetic. These are finite regressions; the all-dimensional statements and asymptotic conclusions rest on the written proof, not on those computations. Errors are raised explicitly, so checks remain active under `python3 -O`.

References

- [1] mxym, entry005, public versions 3–5, pinned commit 31e3d8a37e4a3051a7f5a2535be1c75642e15bcc. <https://github.com/mxym/math/tree/31e3d8a37e4a3051a7f5a2535be1c75642e15bcc/preprints/005-simplex-product-optimum>.
- [2] mxym, *Effective simplex rigidity for the projection cone invariant*, additive supplement to entry005, public source at commit 6785c1c830f8e19e2eb07b0bb89f4d475a8b154a. <https://github.com/mxym/math/blob/6785c1c830f8e19e2eb07b0bb89f4d475a8b154a/notes/quantitative-projection-simplex-stability/paper.md>.
- [3] K. Böröczky, Jr., *The stability of the Rogers–Shephard inequality and of some related inequalities*, Advances in Mathematics 190 (2005), 47–76. Author-hosted manuscript: <https://www.renyi.hu/~carlos/rogerstab.pdf>.

- [4] G. Ambrus and K. J. Böröczky, *Stability results for the volume of random simplices*, American Journal of Mathematics 136 (2014), 833–857. <https://arxiv.org/abs/1005.5024>.
- [5] K. Böröczky, J. M. Machado and J. P. G. Ramos, *Quantitative Stability for Minkowski’s problem*, arXiv:2603.17726v3, 13 May 2026. <https://arxiv.org/abs/2603.17726v3>.
- [6] R. Brandenberg, B. González Merino and F. Grundbacher, *Tight Stability Estimates Near the Simplex and Applications for the Banach–Mazur Distance and Rogers–Shephard-Type Inequalities*, arXiv:2607.27041v2, 14 September 2026. <https://arxiv.org/abs/2607.27041v2>.